

Genome-Wide Expression Analysis Unravels Fluoroalkane Metabolism in *Pseudomonas* sp. Strain 273

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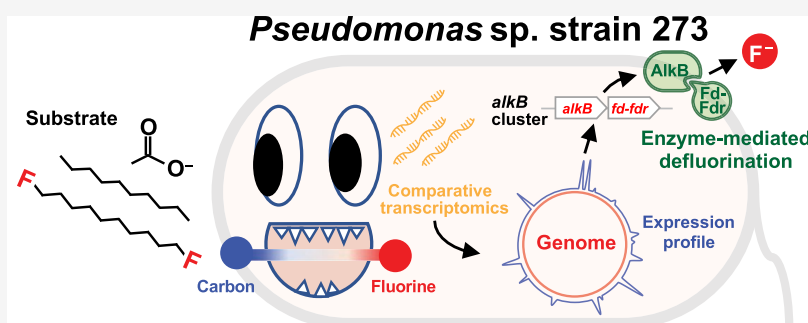
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ABSTRACT: *Pseudomonas* sp. strain 273 grows with medium-chain terminally fluorinated alkanes under oxic conditions, releases fluoride, and synthesizes long-chain fluorofatty acids. To shed light on the genes involved in fluoroalkane metabolism, genome, and transcriptome sequencing of strain 273 grown with 1,10-difluorodecane (DFD), decane, and acetate were performed. Strain 273 harbors three genes encoding putative alkane monooxygenases (AlkB), key enzymes for initiating alkane degradation. Transcripts of *alkB-2* were significantly more abundant in both decane- and DFD-grown cells compared to acetate-grown cells, suggesting AlkB-2 catalyzes the attack on terminal CH₃ and CH₂F groups. Coordinately expressed with *alkB-2* was an adjacent gene encoding a fused ferredoxin–ferredoxin reductase (Fd–Fdr). Phylogenetic analysis distinguished AlkB that couples with fused Fd–Fdr reductases from AlkB with alternate architectures. A gene cluster containing an (S)-2-haloacid dehalogenase (*had*) gene was up-regulated in cells grown with DFD, suggesting a possible role in the removal of the ω -fluorine. Genes involved in long-chain fatty acid biosynthesis were not differentially expressed during growth with acetate, decane, or DFD, suggesting the bacterium’s biosynthetic machinery does not discriminate against monofluoro-fatty acid intermediates. The analysis sheds first light on genes and catalysts involved in the microbial metabolism of fluoroalkanes.

KEYWORDS: fluorinated alkanes, defluorination, genomics, transcriptomics, fluoroalkane metabolism, *Pseudomonas* sp. strain 273

■ INTRODUCTION

Fluorinated chemicals have been utilized in a variety of medical, agricultural, military, industrial, and household applications.^{1–3} As a consequence, fluorinated chemicals have been released into the environment, where some of these chemicals, in particular per- and polyfluoroalkyl substances (PFAS), persist and have emerged as ubiquitously distributed, recalcitrant contaminants.^{4,5} PFAS, comingled with fluorinated alkanes, are components of aqueous film-forming foams (AFFF) used as fire suppressants.^{6,7} Efforts to find microorganisms capable of degrading PFAS have had limited success, although a recent study reported defluorination and degradation in an acidophilic mixed culture dominated by the ammonium-oxidizing bacterium *Acidimicrobium* sp. strain A6.^{8,9} Few studies have explored the degradation of fluorinated alkanes, and the key defluorinating enzyme systems have not been identified.

The strength of the C–F bond has been portrayed as the major hindrance to transform fluorinated organic compounds. Contrary to the belief that enzyme systems cannot cleave C–F bonds, a number of studies have reported the microbial degradation of fluorinated organic compounds.^{10,11} Microbial enzyme systems can break the C–F bond through oxygenolytic,¹² hydrolytic,¹³ reductive,¹⁴ and hydration¹⁵ mechanisms at neutral pH and at room temperature. Despite this progress, the understanding of the taxonomic diversity of microbes and their enzyme systems involved in defluorination dwarfs in

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comparison to the existing knowledge base about microbial dechlorination.

Pseudomonas sp. strain 273 was reported to utilize C₇ to C₁₀ 1-fluoroalkanes and 1,10-difluorodecane (DFD) as the sole sources of carbon and energy for growth under oxic conditions.¹⁶ During growth with fluoroalkanes, this bacterium releases inorganic fluoride, albeit not in stoichiometric amounts, and a fraction of organofluorine is incorporated into cellular phospholipids.¹⁷ Growth experiments determined that the utilization of fluoroalkanes is oxygen-dependent, but the enzyme(s) responsible for defluorination have remained elusive. In this study, we integrated genomic and comparative transcriptomic investigations to identify genes that *Pseudomonas* sp. strain 273 utilizes during growth with DFD.

MATERIALS AND METHODS

Chemicals. DFD (>97%) was custom synthesized (Carbosynth, Newbury, UK). 1-Fluorodecane (purity, >97%) was obtained from SynQuest Labs, Inc. (Alachua, FL, USA). Sodium acetate, chloroform (>99.8%), and sodium fluoride were obtained from Sigma-Aldrich (St. Louis, MO, USA). *n*-Decane (decane) was obtained from Acros Organics (Fair Lawn, NJ, USA). Ethanol (>99.5%) and TRIzol reagent were purchased from Thermo Fisher Scientific (Waltham, MA, USA). All other chemicals used were of analytical reagent grade or higher, unless otherwise specified. Decane and DFD have low aqueous solubilities and indicated are nominal concentrations.

Cultivation of *Pseudomonas* sp. strain 273. The soil isolate *Pseudomonas* sp. strain 273 utilizes acetate, decanoate, decane, and medium-chain-length chloroalkanes and fluoroalkanes as the sole carbon sources.^{16,18} Strain 273 was grown in 160 mL glass serum bottles with 50 mL of phosphate-buffered (pH 7.3), defined mineral salt medium under an air headspace.¹⁶ For the transcriptome study, triplicate vessels for each growth condition, including 25 mM acetate, 5 mM decane, and 5 mM DFD, received 1% (v/v) inocula from strain 273 cultures pregrown on the respective substrates. The vessels were loosely covered with sterile aluminum foil to allow gas exchange. All bottles were shaken at 120 rpm in an upright position at 30 °C. Optical density (OD_{600nm}) readings were recorded in 1 mL culture suspension samples withdrawn at each time point (Figure S1).

DNA Extraction, Genome Sequencing, and Annotation. DNA extraction, genome sequencing, and genome annotation of strain 273 were performed as described.¹⁹ The complete genome of *Pseudomonas* sp. strain 273 is available in GenBank (CP116775) and the JGI IMG/M-ER database (2814122901). The annotation accuracy of the genes of interest was manually verified by blasting gene sequences against the NCBI (<https://www.ncbi.nlm.nih.gov/>) and Uniprot (<https://www.uniprot.org/>) databases. Whole genome average nucleotide identity (ANI) comparisons were conducted using the Microbial Genomes Atlas (MiGA) webserver.^{20,21} The domain analysis of proteins was performed with InterPro.²²

RNA Extraction, Transcriptome Sequencing, and Analysis. Culture suspension samples (5–10 mL containing 4 × 10⁹ to 1 × 10¹⁰ cells) were withdrawn after 24 h from cultures grown with acetate and after 48 h from cultures grown with decane or DFD (Figure S1, Supporting Information). Cells were collected by centrifugation for 15 min at 10,000 g at 4 °C and stored immediately at −80 °C. RNA was extracted

from the cell pellets following an established TRIzol/chloroform extraction protocol, and details are available in the Supporting Information.²³ The individual RNA extracts from triplicate cultures for each of the three different growth conditions (nine samples total) met the quality control standards and were sequenced using the NovaSeq platform (Illumina, San Diego, CA, USA) at the University of Maryland genomics core. Prior to sequencing, rRNA was removed using the NEB Bacterial Reduction Kit (New England Biolabs, Ipswich, MA, USA). The transcript sequences were mapped against the strain 273 genome, and expression differences between growth conditions were assessed following the RNA-seq workflow in the KBase pipeline (<https://www.kbase.us/>), including data trimming, sequence alignment, sequence assembly, and differential expression analysis (see SI for details). The expression level of each gene was normalized to the gene counts and the gene length and is reported as fragments per kilobase of transcript per million mapped reads (FPKM). Two independent alignment methods, HISAT2 and Bowtie2, were applied to the same data set and returned very similar FPKM values after transcript assembly using String-Tie²⁴ (Figure S2, Supporting Information), and the manuscript reports data derived from the Bowtie2 alignments. Differentially expressed genes (DEGs) between growth conditions were determined using DESeq2 (v1.20.0), with the averaged FPKM value above 256 with at least one substrate, a fold change in expression level higher than 2, and a cutoff *p* value of 0.05 in any pairwise comparison.^{25–27} The gene IDs and the FPKM values observed in cultures grown with acetate, decane, and DFD are available in Table S1, Supporting Information.

Reverse Transcription-Quantitative Polymerase Chain Reaction. Reverse transcription-quantitative polymerase chain reaction (RT-qPCR) was applied to enumerate gene transcripts of interest with the goal to corroborate the transcriptome sequencing results. Gene expression data are reported as transcript copies per nanogram of RNA extracted. All RNA extracts were diluted to 2 ng μL^{−1}, and 2 μL aliquots served as templates in RT-qPCR, which used the Power SYBR Green RNA-to-C_T 1-Step Kit (Applied Biosystems, Waltham, MA, USA) and a ViiA 7 Real Time PCR system (Applied Biosystems). Primers specifically amplifying regions of the alkane monooxygenase (*alkB*) genes (i.e., *alkB*-1, IMG gene ID 2814127390; *alkB*-2, 2814128504; and *alkB*-3, 2814127750) were designed using Geneious Prime (v.11.0.6) to meet the following criteria: primer melt temperature (*T*_m) of 57–61 °C and a primer G + C content between 50 and 75 mol % (Table S2, Supporting Information). The specificity of the primers was checked by BLAST search against the strain 273 genome. The double-stranded DNA oligonucleotide sequences carrying the target region and primer binding region of three *alkB* genes were synthesized individually (Integrated DNA Technologies, Coralville, IA, USA) (Table S2, Supporting Information). The RT-qPCR cycle conditions were set as follows: 48 °C for 30 min and then held at 95 °C for 10 min, followed by 40 cycles of 95 °C for 15 s and 60 °C for 1 min. The melt curve analysis performed at the end of each assay included 95 °C for 15 s, 60 °C for 15 s, and 95 °C for 15 s steps. An eight-point standard curve (*R*² > 0.996) for each target gene was generated using serial 10-fold dilutions of the synthetic double-stranded DNA oligonucleotide in three technical replicates. The amplification efficiencies for *alkB*-1, *alkB*-2, and *alkB*-3 were 103.4, 96.0, and 88.1%, respectively, and single peaks on the melt curve plot indicated specific amplification. The expression level of each

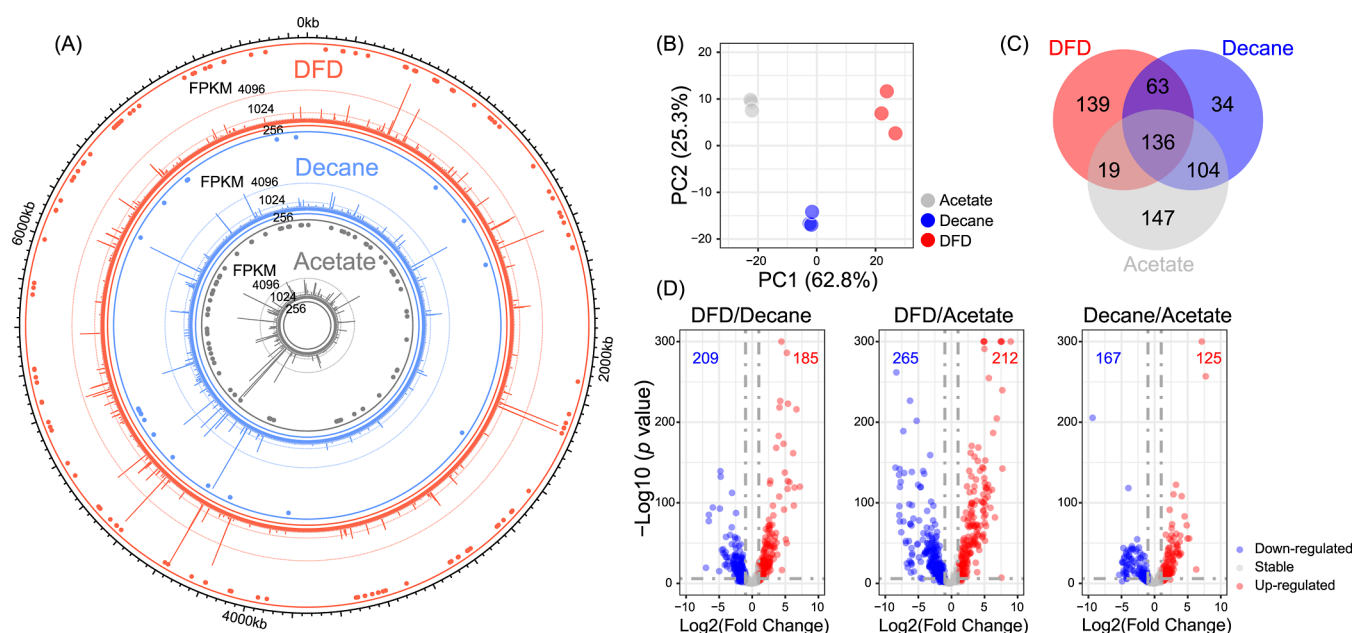


Figure 1. Differential gene expression in strain 273 cells grown with acetate, decane, or DFD as carbon substrates. (A) Global analysis of the strain 273 transcriptome (RNA sequencing). The outermost circle represents the closed genome of strain 273. The circular stick plots in red, blue, and gray represent the FPKM values of genes in strain 273 cells grown with DFD, decane, and acetate, respectively. The dashed circles represent FPKM values of 256, 1024, and 4,096, respectively. The red, blue, and gray dots surrounding the circular stick plots highlight the genes up-regulated in strain 273 cells during growth with the respective substrate. Criteria for up-regulation were a $\text{Log}_2(\text{Fold Change}) > 1$ and a p value < 0.05 in the comparison to the other two substrates. A gene (2814128173) encoding a cold shock protein had an FPKM value exceeding 4882 and is not included in panel A. (B) PCA of all significantly expressed genes with FPKM values > 256 . (C) Venn diagram showing the distribution of significantly expressed genes in cells of strain 273 grown with acetate (gray), decane (blue), and DFD (red), respectively. (D) Volcano plots showing the pairwise comparison of significantly expressed genes between carbon substrates. The expression fold change was determined based on the FPKM values between the indicated carbon substrates. Criteria for calling genes significantly up-regulated and down-regulated included $\text{Log}_2(\text{Fold Change}) > 1$ and $\text{Log}_2(\text{Fold Change}) < -1$, respectively, and p values < 0.05 . In every pairwise comparison, numbers in red and blue font indicate the number of up- and down-regulated genes, respectively. Genes not meeting these criteria were considered to not undergo expression changes with the different carbon substrates. The reported FPKM values represent the averages for each gene determined in triplicate cultures.

gene is determined as the transcript copies normalized to the amount of RNA in the template with the unit of copy number/ng RNA. The extremely low signal obtained in negative control qPCR assays without a reverse transcription step (i.e., reverse transcriptase was omitted) indicated that genomic DNA had been removed. No amplification occurred in the controls without a DNA or RNA template.

Phylogenetic Analysis. A search of the JGI IMG and the Uniprot databases retrieved 51 AlkB homologous proteins from 30 alkane-degrading bacteria, including two taxonomically closely related *Pseudomonas* spp. (Table S3, Supporting Information). Also recovered were 21 amino acid sequences representing (S)-2-haloacid dehalogenase (HAD), haloalkane dehalogenase (HKD), and fluoroacetate dehalogenase (FacD) with experimentally verified dehalogenation activity (Table S4, Supporting Information). Thirteen nucleotide sequences encoding cytochrome P450 (CYP) from nine alkane-degrading bacteria were retrieved from GenBank (Table S5, Supporting Information). Phylogenetic trees were constructed in Geneious Prime (v.11.0.6) using the unweighted pair group method with arithmetic mean (UPGMA) and the Jukes–Cantor model.²⁸

RESULTS

Genome of Strain 273. The genome of strain 273 comprises a single 7,477,410 bp circular chromosome containing 6,758 protein-coding genes (Table S6, Supporting Information). No evidence for the presence of a plasmid was

obtained. Whole genome-based ANI analysis revealed that strain 273 is closely related to *Pseudomonas citronellolis* and *Pseudomonas delhiensis*, with ANI values of 94.6 and 97.8%, respectively. The genome of strain 273 harbors three *alkB* genes (*alkB-1*; *alkB-2*; and *alkB-3*) encoding alkane mono-oxygenases (AlkB), two genes (*had-1* and *had-2*) encoding (S)-2-haloacid dehalogenases (HAD), and one gene encoding a haloalkane dehalogenase (HKD). Complete sets of genes for the β -oxidation pathway and the tricarboxylic acid (TCA) cycle were identified, allowing strain 273 to oxidize acetyl-CoA units generated during alkane catabolism to CO_2 and generate reducing equivalents (i.e., $\text{NADH} + \text{H}^+$, FADH_2). The genome harbors genes encoding cytochrome *c* oxidase, nitrate reductase, nitrite reductase, and nitrous oxide reductase, consistent with growth studies that demonstrated the utilization of oxygen, nitrate, nitrite, and nitrous oxide as respiratory electron acceptors. To identify genes involved in the metabolism of fluorinated alkanes, comparative transcriptome analyses with strain 273 cells grown with acetate, decane, or DFD were performed.

Comparative Transcriptome Analysis Reveals Carbon Substrate-Specific Gene Expression. The global transcriptomic analysis determined 6,755 genes (98.0% of genes on the strain 273 genome) with FPKM values greater than zero (Figure 1A). A total of 642 genes (9.3%) had FPKM values above 256 ($\text{Log}_2\text{FPKM} \geq 8$) and were classified as significantly expressed genes (Tables S1 and S7, Supporting Information).

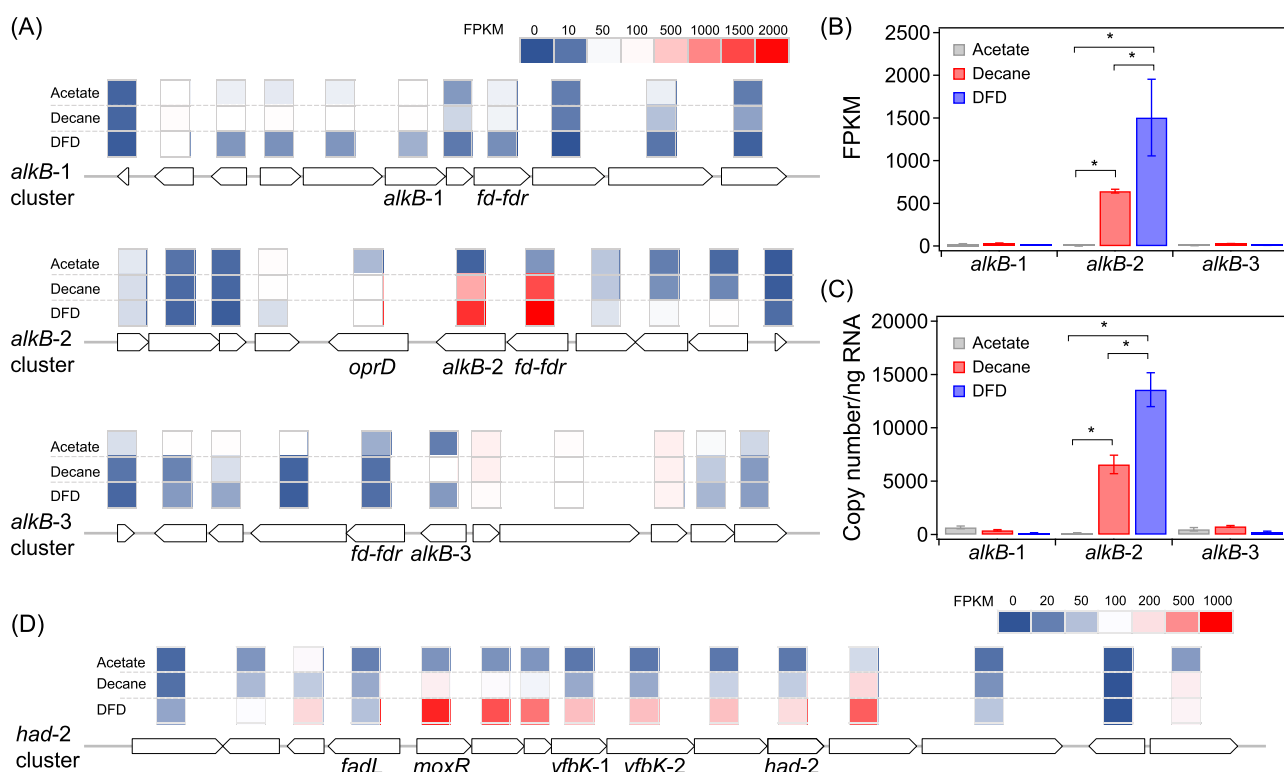


Figure 2. Expression of genes putatively involved in the defluorination fluoroalkanes. (A) Heatmap showing the FPKM values of *alkB-1*, *alkB-2*, *alkB-3*, and surrounding genes during growth with acetate, decane, or DFD. The JGI IMG annotation system was used to infer function and gene abbreviation, with priority given to the KEGG and TIGRFAM classification systems. Genes surrounding *alkB* include *fd-fdr* encoding a fused ferredoxin–ferredoxin reductase and *oprD* encoding outer membrane OprD family porin. Expression of *alkB-1*, *alkB-2*, and *alkB-3* under different growth conditions based on (B) FPKM values and (C) transcript copies per ng of RNA. The data represent the averages of triplicate cultures, and error bars show the standard deviations. Significant differences ($p < 0.001$) of *alkB-2* gene expression determined by a two-tailed *t*-test are marked with asterisks. (D) Heatmap showing the FPKM values of genes adjacent to *had-2* during growth with different carbon substrates. Genes surrounding *had-2* include *fadL* encoding a fatty acid transport protein; *moxR* encoding a MoxR-like ATPase; and *yfbK* encoding a calcium-dependent chloride channel protein.

A principal component analysis (PCA) of these 642 genes separated transcriptomes of cells grown with acetate, decane, and DFD, suggesting specific transcriptional responses to different carbon substrates (Figure 1B). Of these 642 significantly expressed genes, 136 were transcribed under all three growth conditions (Figure 1C), 147 transcripts were exclusively found in cells grown with acetate, and 34 genes were exclusively transcribed in decane-grown cells. During growth with DFD, 357 genes were significantly expressed, with 139 transcripts exclusive to this growth condition.

Of the 642 significantly expressed genes, 533 were determined to be differentially expressed genes (i.e., DEGs) with fold change values > 2 and p values < 0.05 in any pairwise comparison between growth conditions (i.e., decane versus DFD versus acetate as carbon substrates). The volcano plots shown in Figure 1D illustrate the pairwise comparisons of the DEGs in response to the different carbon substrates. The comparison between DFD- and decane-grown cells revealed 185 up-regulated genes in cells grown with DFD and 209 genes showed higher expression in cells grown with decane (Figure 1D). Compared to cells grown with acetate, 212 genes were up-regulated in response to DFD as the carbon substrate (Figure 1D). The pairwise comparison of decane- versus acetate-grown cells revealed 125 up-regulated genes in cells grown with decane (Figure 1D). The genes exclusively up-regulated in response to DFD (159), decane (28), and acetate

(144) are highlighted in Figure 1A as colored dots in the respective substrate.

Of the total of 642 significantly expressed genes, 481 genes have predicted functions according to the Clusters of Orthologous Groups (COG) functional category.²⁹ More than half of the genes were related to the categories translation, ribosomal structure and biogenesis [J], transcription [K], signal transduction mechanisms [T], lipid transport and metabolism [I], energy production and conversion [C], and amino acid transport and metabolism [E] (Table S8, Supporting Information). In the category cell motility [N], cells grown with decane or DFD expressed a higher number of genes (i.e., 20 and 14, respectively) compared to cells grown with acetate (6 genes). When DFD was supplied as the sole carbon source, more genes (44) with a predicted function in the category lipid transport and metabolism [I] were expressed compared to cells grown with acetate (16) or decane (23).

Uptake of Substrates with Low Water Solubility. To access substrates with low water solubility, microorganisms recruit different strategies, including the production of biosurfactants to emulsify the substrate. Common for pseudomonads are rhamnolipids; however, strain 273 does not possess *rhlA* and *rhlB* genes implicated in rhamnolipid biosynthesis.³⁰ Genes encoding the production of the extracellular Pel and Psl polysaccharides and potentially involved in insoluble substrate emulsification were not found in the strain 273 genome.^{31,32} The genome of strain 273

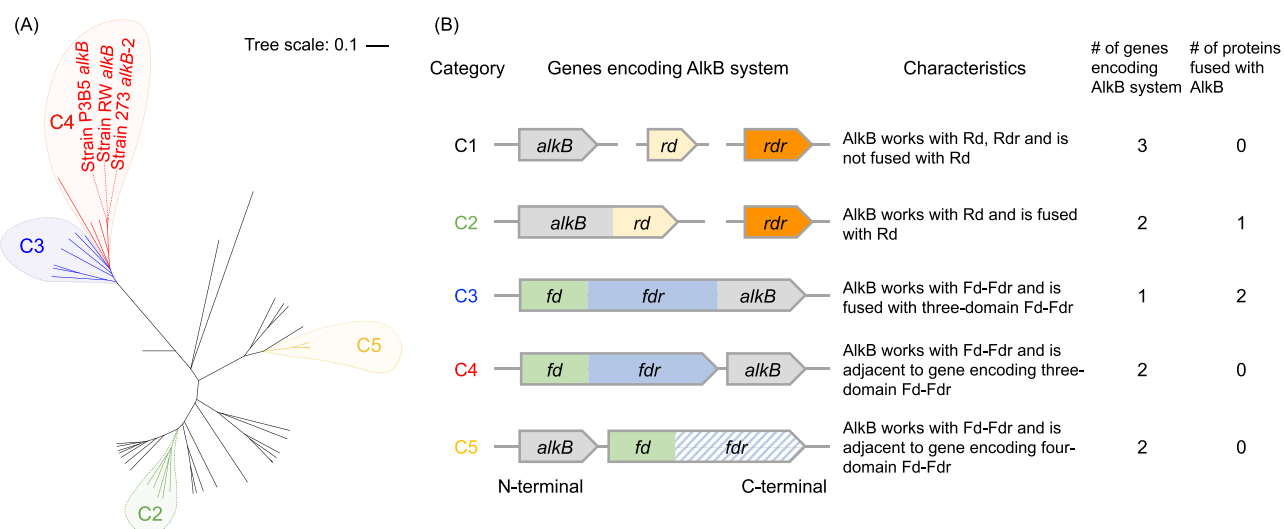


Figure 3. (A) Phylogenetic affiliation of AlkB-1, AlkB-2, and AlkB-3 identified in *Pseudomonas* sp. strain 273 and 48 AlkB enzymes from closely related *Pseudomonas* species and other alkane-degrading bacteria. The tree scale indicates the number of substitutions per amino acid site. The color scheme of the labels C2 to C5 correlates with the gene cluster architecture categories shown in panel B. The C1 category is not labeled in panel A because it is not clear whether all AlkB isolated on the gene cluster work with Rd and Rdr. (B) Features of *alkB* gene clusters and AlkB enzyme systems. Genes/proteins: *alkB*/AlkB, alkane monooxygenase; *fd*/Fd, ferredoxin; *fdr*/Fdr, ferredoxin reductase; *rd*/RD, rubredoxin; *rdr*/RDR, rubredoxin reductase.

encodes a complete pathway for the production of the polysaccharide alginate; however, *algE* (IMG gene ID: 2814125983) and *alg8* (2814125986) were not significantly expressed under all growth conditions (i.e., FPKM values < 7) (Table S1, Supporting Information). An alternative strategy for accessing substrates with limited solubility is active movement toward the alkane droplets (i.e., light, non-aqueous phase liquids).³³ Chemotaxis genes, including genes encoding the methyl-accepting chemotaxis protein MCP (2814129408) and the purine-binding CheW (2814129857), exhibited significant expression in cells grown with decane or DFD. Genes associated with the flagellar apparatus, including *fliE* (2814130009), *fliF* (2814130008), *flgF* (2814130035), and *flhA* (2814129869), were all up-regulated and showed higher expression with alkanes than with acetate (Table S1, Supporting Information).

Genes Putatively Involved in Defluorination. Both AlkB and cytochrome P450 (CYP) can potentially initiate terminal alkane oxidation, which would result in the elimination of one fluorine substituent from DFD.^{34,35} Strain 273 harbors five genes encoding CYP (2814125567, 2814127004, 2814127041, 2814127155, and 2814127435); however, none of the CYP genes was significantly expressed under any tested growth condition (FPKM < 36) (Table S1, Supporting Information). The genome of strain 273 harbors three *alkB* genes, *alkB-1*, *alkB-2*, and *alkB-3*, which share 25 to 31% amino acid sequence identity with each other. No significant expression of *alkB-1* and *alkB-3* occurred under all growth conditions tested (Figure 2A). In contrast, *alkB-2* was significantly expressed in cells grown with decane and DFD, with the highest expression levels observed in cells grown with DFD, and FPKM values of 3 ± 0.2 , 644 ± 22 , and $1,505 \pm 448$ were determined in cells grown with acetate, decane, and DFD, respectively (Figure 2A,B). The expression levels of the three *alkB* genes under different growth conditions were also examined with RT-qPCR (Figure 2C). During growth with acetate, decane, and DFD, 171 ± 10 , $6,581 \pm 865$, and $13,585 \pm 1,595$ *alkB-2* transcripts/ng RNA were measured,

respectively (Figure 2C). Thus, RT-qPCR verified that the expression of *alkB-2* was significantly up-regulated in cells grown with decane or DFD (Figure 2B,C). The quantitative transcript analysis further confirmed that *alkB-1* and *alkB-3* were not significantly expressed under any of the three growth conditions tested (<800 copies/ng RNA) (Figure 2C).

Adjacent to *alkB-2*, we determined an *fd-fdr* gene (2814128505) encoding a fused three-domain ferredoxin–ferredoxin reductase with a [2Fe–2S] iron-sulfur cluster binding domain (pfam00111), an oxidoreductase FAD-binding domain (pfam00970) and an oxidoreductase NAD-binding domain (pfam00175) (Figure S3, Supporting Information). In the COG database, this *fd-fdr* gene is annotated as NAD(P)H-flavin reductase (COG0543). The *fd-fdr* gene followed an expression pattern similar to *alkB-2*, consistent with the coordinated expression of the *alkB* and *fd-fdr* genes. *fd-fdr* genes were also found in the neighborhoods of *alkB-1* and *alkB-3*; however, the gene arrangement differed (Figure S3, Supporting Information). In the *alkB-1* gene cluster, the *fd-fdr* gene (2814127392) encoding a three-domain Fd–Fdr is not adjacent to *alkB-1* and is separated by a 495 bp long gene encoding a hypothetical protein located downstream of *alkB-1* (Figure S3, Supporting Information). In the gene cluster containing *alkB-3*, the *fd-fdr* gene (2814127748) encodes a four-domain protein consisting of a [2Fe–2S] iron-sulfur cluster binding domain (pfam00111), a [2Fe–2S] binding domain (pfam01799), a FAD binding domain (pfam00941), and a CO dehydrogenase flavoprotein C-terminal domain (pfam03450). This protein is annotated as four-domain Fd–Fdr xanthine dehydrogenase, iron–sulfur cluster, and FAD-binding subunit A (XdhA, COG4630) in the COG database and differed from the three-domain Fd–Fdr encoded on the *alkB-2* cluster. The genes neighboring *alkB-1* and *alkB-3*, including the *fd-fdr* genes, all showed low expression levels with FPKM values < 135, similar to the FPKM values of the respective *alkB* genes (Table S1, Supporting Information).

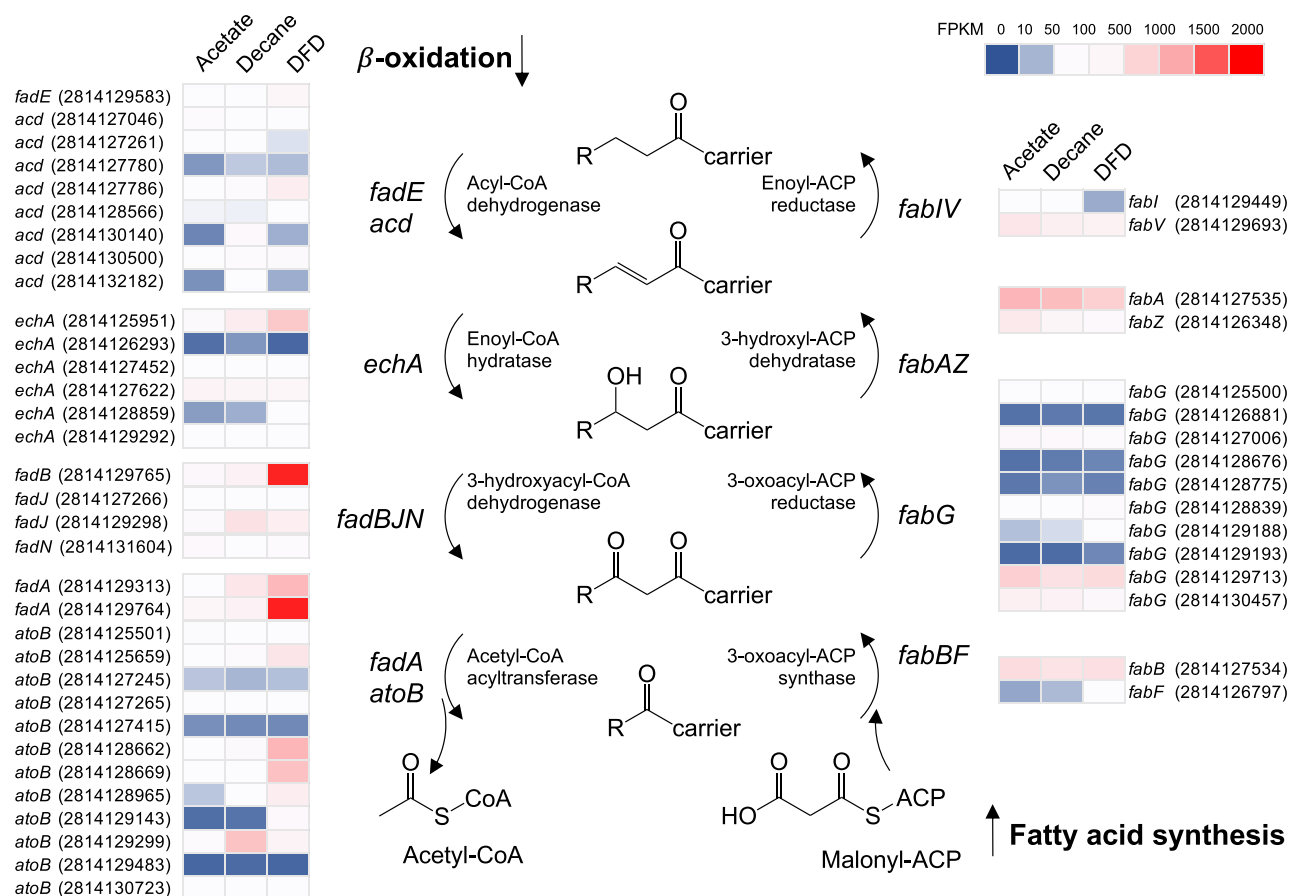


Figure 4. Expression of genes associated with fatty acid metabolism in strain 273 grown with decane, DFD, or acetate. The strain 273 genome harbors a complete set of genes associated with the KEGG fatty acid metabolism pathway (map01212), and the heatmap indicates the expression levels based on FPKM values determined in triplicate cultures. The gene ID numbers are shown in parentheses. The intermediates and enzymes involved in iterative cycles of acyl chain contraction by β -oxidation and extension by fatty acid synthesis are shown. Acyl-CoA dehydrogenase (encoded by *fadE* and *acd*), enoyl-CoA hydratase (*echA*), 3-hydroxyacyl-CoA dehydrogenase (*fadBJN*), and acetyl-CoA acyltransferase (*fadA* and *atoB*) are involved in β -oxidation. 3-Oxoacyl-ACP synthase (encoded by *fabBF*), 3-oxoacyl-ACP reductase (*fabG*), 3-hydroxyl-ACP dehydratase (*fabAZ*), and enoyl-ACP reductase (*fabIV*) are involved in fatty acid biosynthesis. For β -oxidation, the carrier is coenzyme A (CoA), whereas fatty acid synthesis utilizes the acyl-carrier protein (ACP). R represents the (fluorinated) acyl chain.

Haloalkane dehalogenases remove the halogen substituent and convert haloalkanes to primary alcohols, and one putative *hkd* gene (2814131473) was identified on the strain 273 genome. The expression levels of this putative *hkd* gene were low (FPKM < 36) in cells grown with acetate, decane, or DFD. The genome of strain 273 also contains two putative (S)-2-haloacid dehalogenase genes, *had-1* (2814127154) and *had-2* (2814128232), which are candidates responsible for removing the ω -fluorine substituent from fluoroacetate (or fluoroacetyl-CoA). The *had-1* gene was not expressed under any growth condition (FPKM < 5) (Table S1, Supporting Information); however, the *had-2* gene was significantly up-regulated in DFD-grown cells (FPKM 217) compared to acetate- and decane-grown cells (FPKM 29 and 74, respectively) (Figure 2D). The gene cluster containing *had-2* comprises two genes encoding calcium-dependent chloride channel proteins (*yfbK*, 2814128229 and 2814128230), one gene encoding a MoxR-like ATPase (*moxR*, 2814128226), and one gene encoding a fatty acid transport protein (*fadL*, 2814128225). In response to DFD, both *yfbK* and *moxR* genes were upregulated compared to acetate- and decane-grown cells (Table S1, Supporting Information).

Phylogenetic Analysis of *alkB*. The phylogenetic analysis of the three AlkB proteins of strain 273 and 48 AlkB proteins collected from 30 alkane-degrading bacteria demonstrated high inter-species sequence variation with amino acid sequence identities ranging from 15 to 99% (Figure 3A). *alkB* genes have been classified into five categories based on gene cluster architecture. AlkB typically functions in conjunction with rubredoxin (Rd) and rubredoxin reductase (Rdr).³⁵ The gene clusters *alkB*, *rd*, and *rdr* organized in three separate open reading frames (i.e., three genes) represent category 1 (Figure 3B). Category 2 comprises *alkB-rd* fusions encoding an N-terminal alkane monooxygenase with a fused C-terminal rubredoxin. In addition, *alkB* gene clusters encoding protein fusions comprising an N-terminal Fd, an Fdr, and a C-terminal alkane monooxygenase were observed, which represents architectural category 3. The *alkB-2* operon encoding AlkB-2 and a fused N-terminal Fd and C-terminal Fdr represent a new *alkB* gene cluster architecture (category 4). A representative of another novel architecture (category 5) is the *alkB-3* operon of strain 273 consisting of AlkB and an Fd-Fdr fusion protein with four domains. A phylogenetic analysis of *alkB* sequences representing the five architectural categories revealed that the AlkB proteins of categories 3 and 4 have a high amino acid

sequence similarity and form a phylogenetically distinct group that clearly separates from sequences of the other categories (Figure 3A).

Fatty Acid Metabolism in Strain 273. The terminal oxidation of (fluoro)alkanes generates the respective fatty acids, which are catabolized via the β -oxidation pathway. Genes involved in β -oxidation (i.e., the *fad* regulon) encode acyl-CoA dehydrogenase (FadE and Acd), enoyl-CoA hydratase (EchA), 3-hydroxyacyl-CoA dehydrogenase (FadBJN), and acetyl-CoA acyltransferase (FadA).³⁶ In each iterative cycle of β -oxidation, one molecule of acetyl-CoA is released, and the fatty acyl-CoA is shortened by two carbon atoms (Figure 4). Compared to acetate-grown cells, genes involved in β -oxidation, including *echA* (2814125951), *fadJ* (2814129298), and *fadA* (2814129313), were up-regulated in decane- or DFD-grown cells (Figure 4, Table S1, Supporting Information). DFD significantly induced the expression of *fadA* (2814129764) (FPKM 2622) and *fadB* (2814129765) (FPKM 2593) in strain 273 while transcript abundances in cells grown with decane and acetate had FPKM values below 150. The strain 273 genome harbors 12 *atoB* gene homologs encoding acetyl-CoA acetyltransferase involved in releasing acetyl-CoA in the last step of iterative β -oxidation cycles. Three gene homologs, including *atoB* (2814125659) (FPKM 297), *atoB* (2814128662) (FPKM 849), and *atoB* (2814128669) (FPKM 722), were significantly expressed in cells grown with DFD but not under the other two growth conditions (FPKM < 54). In cells grown with decane, the homolog *atoB* (2814129299) was significantly up-regulated (FPKM 703) but showed lower expression in cells grown with DFD (FPKM 124) or acetate (FPKM 41).

The biosynthesis of glycerophospholipid for cellular membrane assembly requires iterative cycles of C₂ unit additions by condensing a malonyl-ACP with an acyl-ACP molecule, leading to the extension of the fatty acyl chain by two carbon atoms per cycle. In contrast to genes involved in β -oxidation, the genes encoding enzymes involved in fatty acid synthesis, including 3-oxoacyl-ACP synthase (*fabB*) (2814127534), 3-oxoacyl-ACP reductase (*fabG*) (2814129713), and 3-hydroxyacyl-ACP dehydratase (*fabA*) (2814127535), were not differentially expressed in response to different carbon substrates and had fold change numbers of less than two between any pairwise comparison.

DISCUSSION

Molecular Underpinning of (Fluoro)alkane Oxidation. Previous work demonstrated that strain 273 utilizes decane, 1,10-dichlorodecane, and DFD as growth substrates and suggested that the same enzyme system is involved in the initial attack on the terminal carbon and the release of inorganic halides.^{16,18} The genome of strain 273 harbors one putative *hkd* gene encoding a haloalkane dehalogenase, five genes encoding P450-type monooxygenases (CYP), and three *alkB* genes encoding alkane monooxygenases, all of which could potentially play a role in the initial attack on DFD. The haloalkane dehalogenase in *Rhodococcus rhodochrous* strain NCIMB 13064 was shown to dehalogenate monochloroalkanes with carbon chain length ranging between C₃ and C₁₀ but not fluoroalkanes.³⁷ Most alkane-oxidizing CYPs belong to the CYP153 subfamily,³⁸ but the five CYPs in strain 273 do not belong to this group (Figure S4, Supporting Information). Consistent with these observations, strain 273 did not express *hkd* and CYP genes during growth with acetate, decane, or

DFD. The transcript analysis suggested that AlkB-2, encoded by one of three *alkB* genes on the strain 273 genome, initiates the attack on both decane and DFD.

AlkB-Mediated Defluorination. AlkB-type alkane hydroxylases are integral membrane non-heme diiron monooxygenases. Based on current understanding of AlkB-catalyzed reactions, AlkB-2 is predicted to generate an unstable geminal fluorohydrin, which undergoes spontaneous intracellular elimination with the concomitant release of inorganic fluoride and a proton.¹⁶ The resulting ω -fluoro-aldehyde is oxidized to the ω -fluoro-carboxylate, which is channeled into the β -oxidation pathway.^{39,40}

Metabolomic and lipidomic investigations revealed that strain 273 incorporates organofluorine into its membrane phospholipids during growth with 1-fluorodecane and DFD.¹⁷ Based on the observation that the percentage of fluorinated lipids in strain 273 cells was ~ 11 -fold higher in DFD versus 1-fluorodecane grown cells, it was suggested that the initial attack preferentially occurs at the fluorinated carbon.¹⁶ Apparently, AlkB-2 has a preference for the fluoromethyl group rather than the methyl group. This potentially distinguishing feature of AlkB-2 to characterized AlkB enzymes warrants further investigation to reveal structural characteristics that favor activity toward fluorinated alkanes. The expression analysis detected more *alkB*-2 transcripts in DFD- versus decane-grown cultures (Figure 2B,C), suggesting that DFD may be a stronger inducer of the *alkB*-2 operon than decane; however, time-series expression analysis would be needed to verify this observation.

Canonical AlkB systems are protein complexes consisting of the catalytically active alkane monooxygenase AlkB and two electron transfer proteins, rubredoxin (Rd) and rubredoxin reductase (Rdr) (Figure 3B, category C1).^{34,41} More recent work has reported alternate architectures consisting of AlkB-Rd (C2) and Fd-Fdr-AlkB (C3) fusion proteins (Figure 3B).^{35,42,43} The observed coordinated expression of *alkB*-2 and *fd-fdr* transcripts revealed a new AlkB system architecture (C4) comprising AlkB and a fused Fd-Fdr protein involved in the initial attack on both decane and DFD. Complexes consisting of AlkB, Fd, and Fdr are not well studied in comparison to category C1 representatives, and the biological consequences of the different architectures for AlkB activity have not been experimentally explored. Fd ($E_0' = -0.39$ V) has a lower redox potential than Rd ($E_0' = -0.06$ V) and is more thermodynamically favorable in activating oxygen and initiating the electrophilic attack on the terminal carbon.⁴⁴ In addition, the Fd-Fdr fusion may allow higher electron transfer rates, simplify regulatory circuits, or expand the substrate range of AlkB enzyme systems.^{43,45} Further experimental work is warranted to explore the role of Fd-Fdr fusion proteins and the requirement for Fd (rather than Rd) for an AlkB system capable of attacking a terminal carbon with a fluorine substitution.

Fluoroalkane Metabolism. In decane- and DFD-grown cells, genes associated with the assembly of the flagellar motor (*fliEF*, *flgF*, and *flhA*) were upregulated, presumably in response to growth with water-immiscible substrates (Figure 5). Motility allows the cells to actively move towards insoluble hydrocarbon droplets to access the substrate.^{46,47} Some microorganisms generate biosurfactants, such as rhamnolipids and polysaccharides, to emulsify immiscible substrates and enhance substrate accessibility.^{32,48} However, the genomic and transcriptomic analyses provided no evidence for the production of biosurfactants, suggesting strain 273 relies on

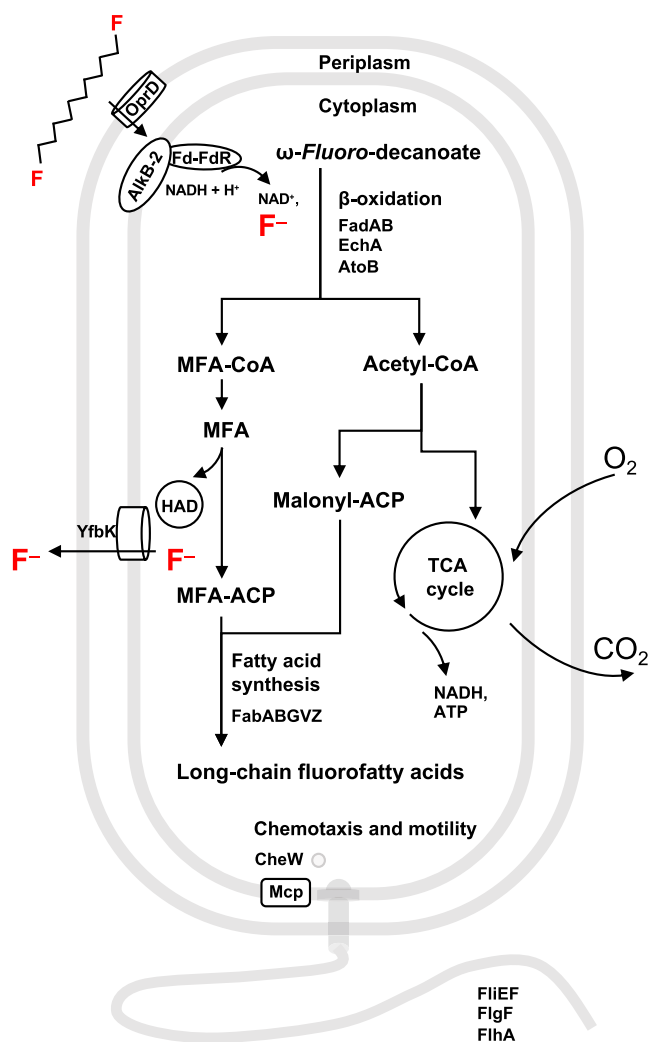


Figure 5. Proposed DFD metabolic pathway in strain 273 based on genome annotation and transcriptional expression profiles. The enzymes and metabolites listed are associated with the uptake and catabolism of fluorinated alkanes, and the channeling of intermediates into anabolic pathways (e.g., long-chain fatty acid biosynthesis). Abbreviations: TCA cycle, tricarboxylic acid cycle; MCP, methyl-accepting chemotaxis protein; CheW, purine-binding chemotaxis protein; FliE, flagellar hook-basal body complex protein; FliF, flagellar M-ring protein; FlgF, flagellar basal-body rod protein; FlhA, flagellar biosynthesis protein.

direct contact between substrate droplets and motile cells to access substrates with limited aqueous phase solubility (i.e., decane, DFD), similar to what has been observed during hexadecane utilization by various *Pseudomonas* spp.^{46,49}

Proteins of the OprD family form channels to facilitate the uptake of a variety of substrates in *Pseudomonas aeruginosa*.⁵⁰ Located adjacent to *alkB-2* is *oprD* (2814128503), which encodes an OprD family protein putatively involved in the import of DFD (Figure 5). AlkB-2, an integral inner membrane protein, then attacks DFD, leading to the release of inorganic fluoride from one of the two terminal fluoromethyl groups. Subsequent action of an aldehyde dehydrogenase yields ω -fluoro-decanoate, which feeds into the β -oxidation pathway. The elevated expression levels of β -oxidation pathway genes (i.e., *fadA* and *echA*) indicate that this pathway is operational during growth with decane and DFD but not during growth with acetate. Three *atoB* gene homologs encoding acetyl-CoA

C-acetyltransferase catalyzing the conversion of 2 acetyl-CoA to CoA + acetoacetyl-CoA were differentially expressed in cells grown with decane and DFD. *atoB* (2814128662) and *atoB* (2814128669) homologs were up-regulated in cells grown with DFD, while *atoB* (2814129299) was up-regulated in cells grown with decane. This observation suggests strain 273 utilizes different AtoB enzymes for metabolizing fluorinated versus non-fluorinated substrates via the β -oxidation pathway. β -Oxidation of one ω -fluoro-decanoate molecule (derived from one DFD molecule) would result in the formation of one inorganic fluoride, four acetyl-CoA, and one monofluoroacetate- (MFA-) CoA. Part of the acetyl-CoA can be metabolized through the TCA cycle, generating reducing equivalents (e.g., NADH + H⁺, FADH₂). A fraction of the acetyl-CoA can be transformed to malonyl-ACP for anabolic purposes such as long-chain fatty acid biosynthesis. In DFD-grown cells, *had-2* and neighboring genes encoding a fatty acid transport protein FadL (2814128225), a MoxR-like ATPase (2814128226), and two YfbK-type anion channel proteins (2814128229 and 2814128230) were up-regulated, suggesting a possible involvement of an (S)-2-haloacid dehalogenase in the defluorination of fluorinated fatty acid catabolites such as MFA-CoA. MFA can be formed from MFA-CoA following thioester hydrolysis and converted by HAD-2 to the central metabolite glycolate and inorganic fluoride. HAD-2 affiliates with HADs that have demonstrated activity toward fluoroacetate and chloroacetate (Figure S5, Supporting Information),^{51–53} however, functional prediction of HADs based on sequence analysis has limitations,⁵¹ and further studies are required to test if HAD-2 encoded on the strain 273 genome has defluorination activity. A prior study reported that strain 273 co-metabolizes MFA with decane as the primary substrate,¹⁶ supporting a role of *had-2* in defluorination.

The CLC family of fluoride ion transporters and channels (CLC^F) and the fluoride ion transporter Fluc-F⁻ have been reported to transport potentially toxic fluoride ions outside of the cell.^{54,55} For example, Fluc-F⁻ enables *Pseudomonas putida* strain KT2440 to tolerate high fluoride concentrations.^{56,57} Two *crb* genes (2814130449 and 2814128936) encoding Fluc-F⁻ transporters were not significantly expressed (FPKM < 70) in strain 273 cells grown with DFD. The *yfbK* genes (2814128229 and 2814128230) encoding YfbK-type anion channel proteins, located on the same gene cluster as *had-2*, were significantly expressed and up-regulated in cells grown with DFD compared with decane- and acetate-grown cells, implicating that they might be involved with fluoride export in strain 273.

A previous study demonstrated the synthesis of long-chain fatty acids with a single fluorine substitution in strain 273 cells grown with DFD (or 1-fluorodecane),¹⁷ suggesting that a small portion of the MFA-CoA is used for anabolic reactions. The genes associated with fatty acid synthesis, including *fabABGVZ*, were not differentially expressed in response to growth with acetate, decane, or DFD, suggesting the same enzymatic machinery processes fluorinated and non-fluorinated fatty acid biosynthesis intermediates.¹⁷ This enzyme promiscuity can explain the formation of fluorinated long-chain fatty acids in strain 273 grown with fluorinated alkanes.

Distribution of AlkB with Potential Defluorination Activity. *alkB* genes are highly diverse, and a recent metagenome-based study suggested that an even broader *alkB* diversity exists in not-yet-cultivated microorganisms.⁵⁸ The *alkB-2* of strain 273 has homologs in *Pseudomonas* sp.

strain RW407 (A0A2V2SWG1) and *Pseudomonas citronellolis* strain P3B5 (2688823651) with amino acid identities of 99.2 and 92.1%, respectively (Figure 3A). Comparative genome analysis revealed that the three highly similar *alkB* genes of strains 273, RW407, and P3B5 share the same gene cluster architecture (i.e., *alkB* adjacent to a fused *fd*–*fdr* gene; Figure 3B, architecture C4). Both strain RW407 and strain P3B5 were experimentally demonstrated to defluorinate DFD (data not shown), suggesting *alkB*-2 is a potential biomarker for fluoroalkane degradation. All three DFD degraders (i.e., strains 273, RW407, and P3B5) share the C4 architecture, and future research should explore if this architecture is required and could thus serve as a marker for initiating the attack on terminally fluorinated alkanes. *Pseudomonas* sp. strain RW407 was isolated from polluted sediment of the Elbe River in Germany,⁵⁹ *Pseudomonas citronellolis* strain P3B5 was isolated from the basil plant phyllosphere,⁶⁰ and strain 273 was obtained from garden soil collected in the Stuttgart area, Germany.^{18,59} The high sequence similarity between *alkB*-2 of strain 273 and the *alkB* genes of strain RW407 and strain P3B5 could indicate horizontal gene transfer, an observation supported by a recent study that analyzed *alkB* genes in a genomic context and concluded that horizontal gene transfer events contribute substantially to the dissemination of *alkB* genes.⁶¹

The comparative transcriptome analysis revealed a unique AlkB alkane monooxygenase with a novel AlkB Fd-Rd architecture involved in the initial attack on decane and DFD. AlkB-2 is the first enzyme of this type implicated in oxygenolytic defluorination, offering opportunities to investigate features distinguishing AlkB enzymes that attack a terminal fluoromethyl group. The cluster architecture suggests that genetic features, including *fd*–*fdr* fusions adjacent to *alkB*, might determine oxygenolytic defluorination capability. As more *alkB* gene clusters with the ability to attack terminally fluorinated alkanes are identified and their sequence space better constrained, distinguishing characteristics, or lack thereof, will emerge. If distinguishing sequence characteristics correlate with the ability to attack fluoromethyl groups, assays for measuring defluorination biomarkers can be developed with potential utility for in situ monitoring of defluorination activity. Such information can prove useful at sites undergoing PFAS remediation, where physical-chemical treatments can increase the formation of fluorinated alkanes with a low degree of fluorination and are susceptible to microbial metabolism.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.3c03855>.

Growth curves of strain 273 with acetate, decane, or DFD as carbon substrates; comparison of FPKM values for genes determined with two different alignment tools, HISAT2 and Bowtie2; conserved gene cluster architecture and domain components of each AlkB encoded on the genome of strain 273; nucleotide sequence-based phylogenetic tree of cytochrome P450 genes; amino acid sequence-based phylogenetic tree of (S)-2-haloacid dehalogenase, fluoroacetate dehalogenase, and haloalkane dehalogenase; list of expressed genes in strain 273 cells grown with acetate, decane, or DFD; primers and oligonucleotide standards used for RT-

qPCR enumeration of *alkB*-1, *alkB*-2, and *alkB*-3 transcripts; alkane monooxygenase homologs present in *Pseudomonas* species phylogenetically related to strain 273 and other alkane-degrading bacteria; information about HAD, FAcD, and HKD retrieved from the Uniprot and the JGI IMG databases; information about cytochrome P450 genes encoded on the genomes of select alkane-degrading bacteria; features of the *Pseudomonas* sp. strain 273 genome; the gene expression distributions in strain 273 at different FPKM values; and COG-based functional classification and number of significantly expressed genes in strain 273 cells grown with different carbon substrates (PDF)

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Notes

The authors declare no competing financial interest.

The genome of strain 273 is available in GenBank under accession number CP116775, with BioProject and BioSample accession numbers of PRJNA926353 and SAMN32870201, respectively. The raw reads and the complete genome sequence of *Pseudomonas* sp. strain 273 have been deposited in GenBank (CP116775) and the Sequence Read Archive (SRR24356093). The JGI-IMG annotated genome is available under the name “*Pseudomonas* sp. strain 273” with IMG genome ID 2814122901. The transcriptomic results (i.e., FPKM value for every gene) reported in this study are available in the Table S1, Supporting Information.

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